Pediatric Hypotension / Shock

PM-03 | Pediatric Medical

Quick flow

01 Recognize Shock [EMT / AEMT / Paramedic]

Assess mental status, central/peripheral pulses, capillary refill, skin temperature/color, work of breathing, BP trends, history, and likely cause. Hypotension is a late sign.

02 Airway + Oxygenation [EMT / AEMT / Paramedic]

Support ventilation, control external hemorrhage, keep warm, check glucose, and monitor ECG/SpO_2. AEMT/Paramedic: obtain rapid IV/IO without delaying care.

03 Identify Shock Type [EMT / AEMT / Paramedic]

Hypovolemic/hemorrhagic • distributive/septic • cardiogenic • obstructive. Treat the likely cause while supporting perfusion.

04 Fluid Strategy [AEMT / Paramedic]

Nonhemorrhagic hypovolemic/distributive: isotonic crystalloid 10-20 mL/kg aliquots with reassessment after every bolus. Cardiogenic: 5-10 mL/kg cautiously with early Medical Control. Suspected traumatic hemorrhage: control bleeding, expedite transport, and obtain Medical Control direction; do not apply the medical/septic bolus target indiscriminately.

05 Fluid-Refractory / Obstructive? [Paramedic]

For septic shock unresponsive to fluid, begin Medical-Control-directed epinephrine or norepinephrine infusion. Decompress tension pneumothorax and treat other obstructive causes when indicated.

06 Rapid Transport [EMT / AEMT / Paramedic]

Reassess perfusion, lungs, mental status, glucose, and BP after every intervention; notify a pediatric-capable destination early.

Overview

- Compensated pediatric shock may precede hypotension; recognize abnormal perfusion before blood pressure falls.
- Classify shock as hypovolemic/hemorrhagic, distributive, cardiogenic, or obstructive because fluid strategy and definitive treatment differ.

Provider-Level Actions

Use a length- or weight-based pediatric system and reassess after every intervention.

EMT

- Support airway/ventilation, control external hemorrhage, keep the child warm, assess glucose, apply monitoring when available, and expedite transport.
- Identify early compensated-shock findings: tachycardia, weak pulses, delayed refill, cool/mottled skin, altered behavior, reduced urine output/poor intake, and worsening work of breathing.

AEMT

- Perform all EMT care plus rapid IV/IO access and shock-type-specific isotonic-crystalloid aliquots with reassessment after every dose. The medical/septic aliquot instructions do not establish a pediatric traumatic-hemorrhage order; obtain Medical Control direction for that branch.
- Do not delay treatment for repeated IV attempts; use IO promptly when vascular access is not quickly available in critical shock.

Paramedic

- Perform all prior care plus cardiac monitoring, needle decompression for tension pneumothorax, and Medical-Control-directed vasoactive infusion for fluid-refractory septic shock.
- Use epinephrine or norepinephrine as the initial vasoactive infusion for fluid-refractory septic shock when authorized. For suspected adrenal crisis, give the patient's prescribed emergency steroid or Medical-Control-directed equivalent.

Indications

- Pediatric patient with poor perfusion, altered mental status, weak pulses, delayed capillary refill, cool/mottled skin, or age-defined hypotension.

Treatment steps

- Do not use a single shock-index shortcut as the sole determinant of pediatric shock. Use age-adjusted vital signs, perfusion findings, and clinical trends.
- Give isotonic crystalloid in 10- or 20-mL/kg aliquots for nonhemorrhagic hypovolemic/distributive shock and reassess after every bolus for fluid responsiveness and volume overload.
- Avoid routine 20-mL/kg boluses in suspected cardiogenic shock; use 5-10 mL/kg cautiously, reassess, and open PC-03 Pediatric Pulmonary Edema / CHF.
- For fluid-refractory septic shock, begin Medical-Control-directed epinephrine or norepinephrine infusion; do not use routine corticosteroids unless adrenal crisis is suspected or Medical Control directs otherwise.
- For suspected traumatic hemorrhage, control bleeding, prevent hypothermia, and expedite pediatric-capable trauma care. Use a separately approved pediatric traumatic-hemorrhage order or online Medical Control for fluid selection, aliquot, target, and reassessment limits; do not automatically apply medical/septic-shock boluses or cumulative targets. Blood products require an authorized, supported transfusion program.
- For suspected TBI, prevent hypotension and do not use permissive hypotension. The generic hypotension recognition thresholds below are not TBI perfusion targets; obtain Medical Control direction until the pediatric TBI numeric target table and age boundaries are approved.
- Treat the cause: control hemorrhage, give epinephrine for anaphylaxis within the indicated provider scope, obtain Paramedic-only decompression for tension pneumothorax when indicated, and use the appropriate cardiac or airway pathway.

Medications

Isotonic crystalloid - nonhemorrhagic hypovolemic/distributive shock

10-20 mL/kg IV/IO aliquots; reassess perfusion, lungs, hepatomegaly, and work of breathing after every bolus. Further fluid is individualized with Medical Control.

Isotonic crystalloid - cardiogenic shock

5-10 mL/kg IV/IO cautiously with early Medical Control; reassess after each aliquot

Epinephrine or norepinephrine infusion

Medical-Control-directed for fluid-refractory septic shock; verify pediatric formulation and dose in the current medication reference

Emergency corticosteroid

For known/suspected adrenal crisis: use the patient's prescribed emergency plan or Medical-Control-directed protocol equivalent

Warnings / precautions

- Do not use medical/septic-shock fluid targets indiscriminately for traumatic hemorrhage. Do not substitute generic pediatric hypotension recognition thresholds for TBI-specific perfusion targets.

- Hypotension is late and ominous in pediatric shock: neonate SBP <60, 1 month-<1 year <70, age 1-9 <70 + (2 × age), age ≥10 <90 mmHg.
- Avoid routine 20-mL/kg boluses in suspected cardiogenic shock; pulmonary edema may worsen rapidly.
- Reassess after every fluid bolus for fluid responsiveness and signs of volume overload.
- Do not use routine corticosteroids for septic shock unless adrenal crisis is suspected or Medical Control directs otherwise.

Clinical pearls

- Early pediatric shock is recognized by perfusion and behavior changes before hypotension develops.
- Epinephrine or norepinephrine is preferred for fluid-refractory pediatric septic shock when authorized and available.

Related pathways / contacts

TB-06 Multiple Trauma - Open for hemorrhage control, trauma priorities, and destination guidance; pediatric traumatic-hemorrhage fluid parameters still require Medical Control.

TB-05 Head Trauma - Open for TBI oxygenation and ventilation safeguards; pediatric TBI blood-pressure targets are Medical Control-directed.

AR-05 Pediatric Airway - Open for airway and ventilation support.

PC-03 Pediatric Pulmonary Edema / CHF - Open for suspected cardiogenic shock or pulmonary edema.

PM-01 Pediatric Anaphylaxis - Open for anaphylactic shock.

Medication Reference - Open the current Claiborne medication reference.

Source and references

Claiborne County EMS PM-03 Pediatric Hypotension / Shock protocol, approved October 26, 2026. | Surviving Sepsis Campaign: International Guidelines for the Management of Septic Shock and Sepsis-associated Organ Dysfunction in Children. | Endocrine Society clinical guidance for primary adrenal insufficiency and congenital adrenal hyperplasia.

Revision: 2026-10-30 | Last verified: 2026-10-30